



Medicine

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FROM THE PRESIDENT

Hello readers! This is Danu Kim, the President of KISJ Medical Club. This year, KISJ Medical Club embarks on a new project, publishing the KISJ Medical Club Magazine, *Medicine*. *Medicine* will cover various topics from the most recent medical research to complex medical topics that are not easily accessible to regular high school students. *Medicine* stands as the medium for the KISJ Medical Club to enhance medical knowledge of the KISJ community and provide interesting medical insights.

This month's theme is Neuroscience, a topic full of excitement. With this magazine, you will learn so much about the brain and neuroscience breakthroughs, from ADHD management to fetal brain development. Now, I will stop talking and let you read. Please enjoy!

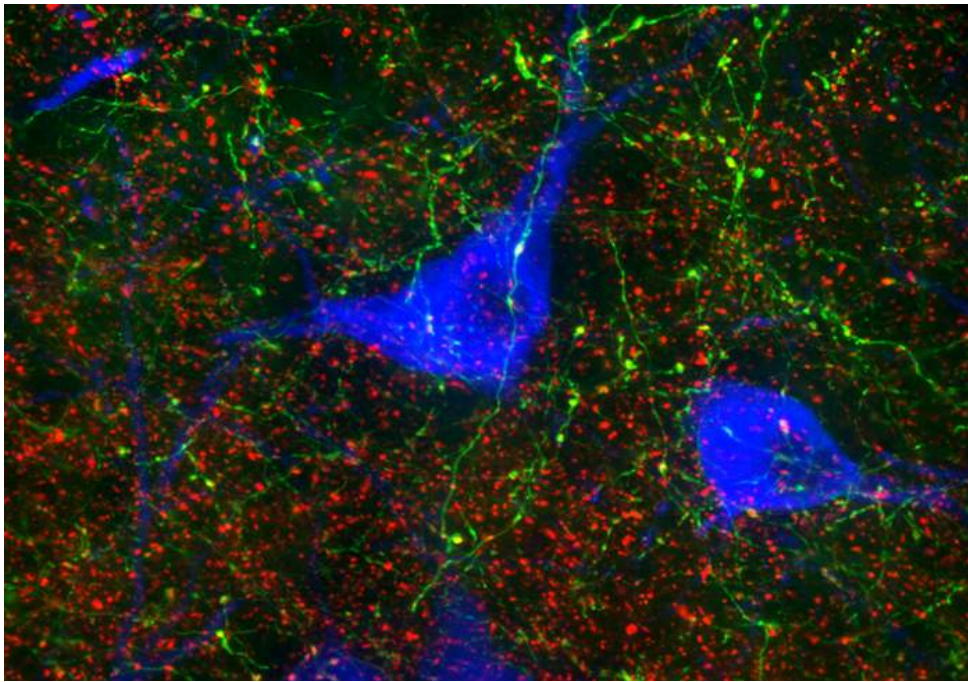
- Danu Kim, President of KISJ Medical Club

The Evolution of ADHD Management: Beyond the Stimulant Paradigm

BY HYUNJUN (HENRY) KIM

The Stimulant Hegemony and Its Limits

Since the 1960s, stimulants like methylphenidate (Ritalin) and amphetamines (Adderall) have been the clinical gold standard for ADHD. By increasing dopamine and noradrenaline levels, these drugs offer high immediate efficacy. However, as of 2026, the medical community faces a critical gap: roughly 30% of patients are either non-responsive or cannot tolerate side effects like cardiac strain, insomnia, and mood volatility. Furthermore, it remains debated whether stimulants truly improve long-term academic quality or merely provide the impression of better performance.



(Paul Hatini, Lab of Kathryn Commons, 2018)

prefrontal cortex. This prevents neural circuits from "going offline" during stress, strengthening focus without the addictive potential of controlled substances.

New Neurological Framework

Modern research identifies ADHD as a structural condition involving underdeveloped gray matter in the frontal lobe. Emerging therapies look beyond dopamine to a broader neurotransmitter landscape - including serotonin, GABA, and glutamate - to move past the "one-size-fits-all" approach. A breakthrough by Dr. Amy Arnsten (Yale University) highlights the potential of $\alpha 2A$ agonists (e.g., Guanfacine). Unlike stimulants that flood synapses with chemicals, these agents work "mechanically" by closing potassium channels in the

The Rise of Diversified Therapeutics

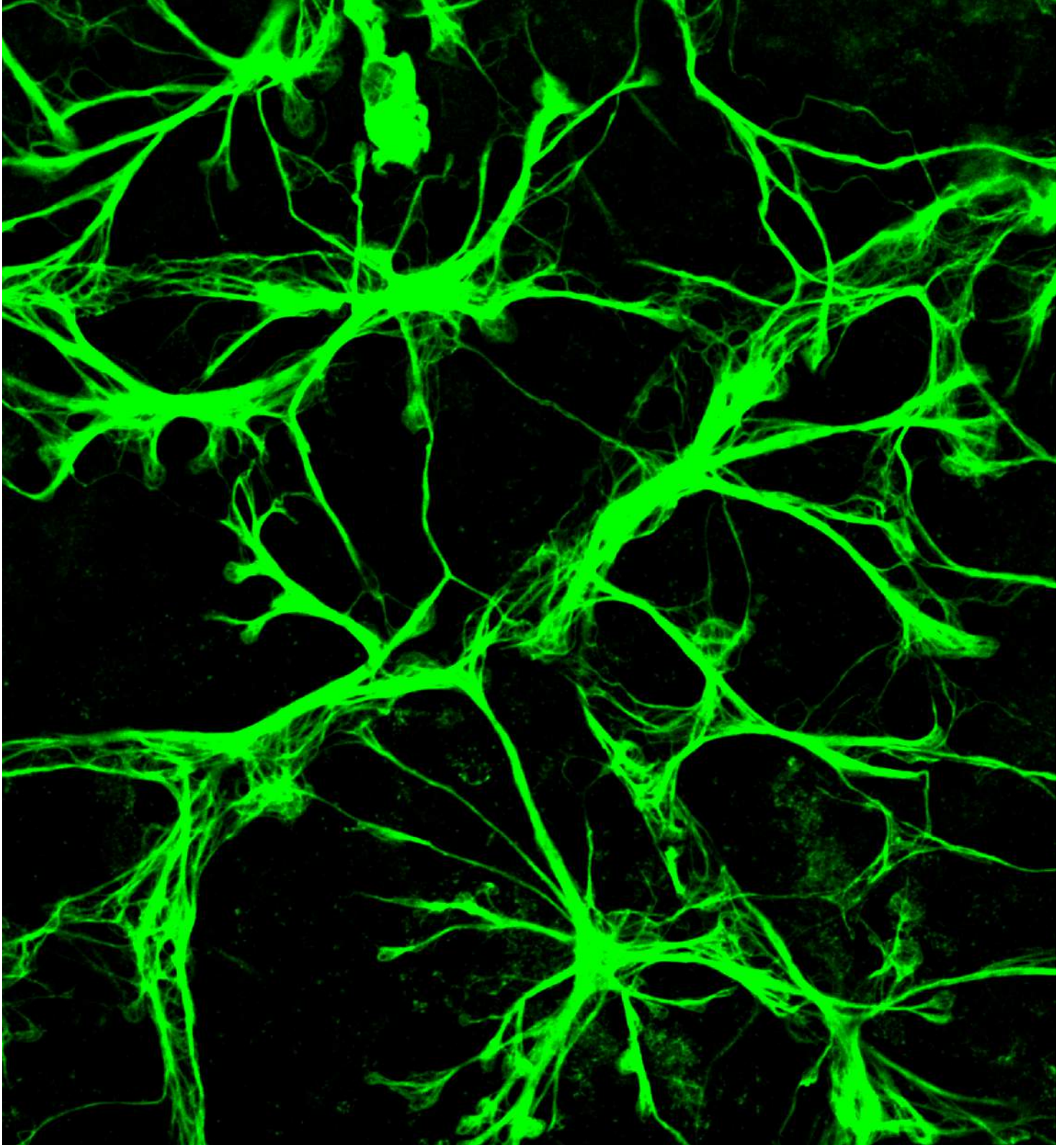
Recent clinical trials, synthesized by Nicola Jones, outline three major non-stimulant frontiers:

- **Novel Pharmacology:** Drugs like Centanafadine (a triple reuptake inhibitor) aim to balance mood and focus simultaneously, offering a "creative" focus rather than the rigid "tunnel vision" associated with Ritalin.
- **Digital Pharmacy:** The FDA-approved EndeavorRx video game targets neural pathways for attention. While less potent than pills, it provides a side-effect-free modality for pediatric populations.

Neuromodulation: Transcranial Direct Current Stimulation (tDCS) has shown significant promise. Recent 4-week trials in adults demonstrated that low-level electrical "priming" of the brain can occasionally rival the efficacy of traditional stimulants.

Conclusion: Toward Precision Psychiatry

The shift toward non-stimulant treatments signals the era of Precision Psychiatry. By repurposing blood pressure medications and integrating digital tools, the field is moving toward a future where treatment is tailored to a patient's specific genetic and neurological profile. For the medical community, this diversification reduces reliance on controlled substances and offers safer, sustainable alternatives for patients with complex medical histories.



(Nikita Bagaev, Lab of Petr Baranov, 2025)

Jones, N. (2026). ADHD treatments move beyond stimulants. *Nature*.

Harvard Brain Science Initiative. (n.d.). Beauty of the brain. Harvard University. <https://brain.harvard.edu/beauty-of-the-brain/>

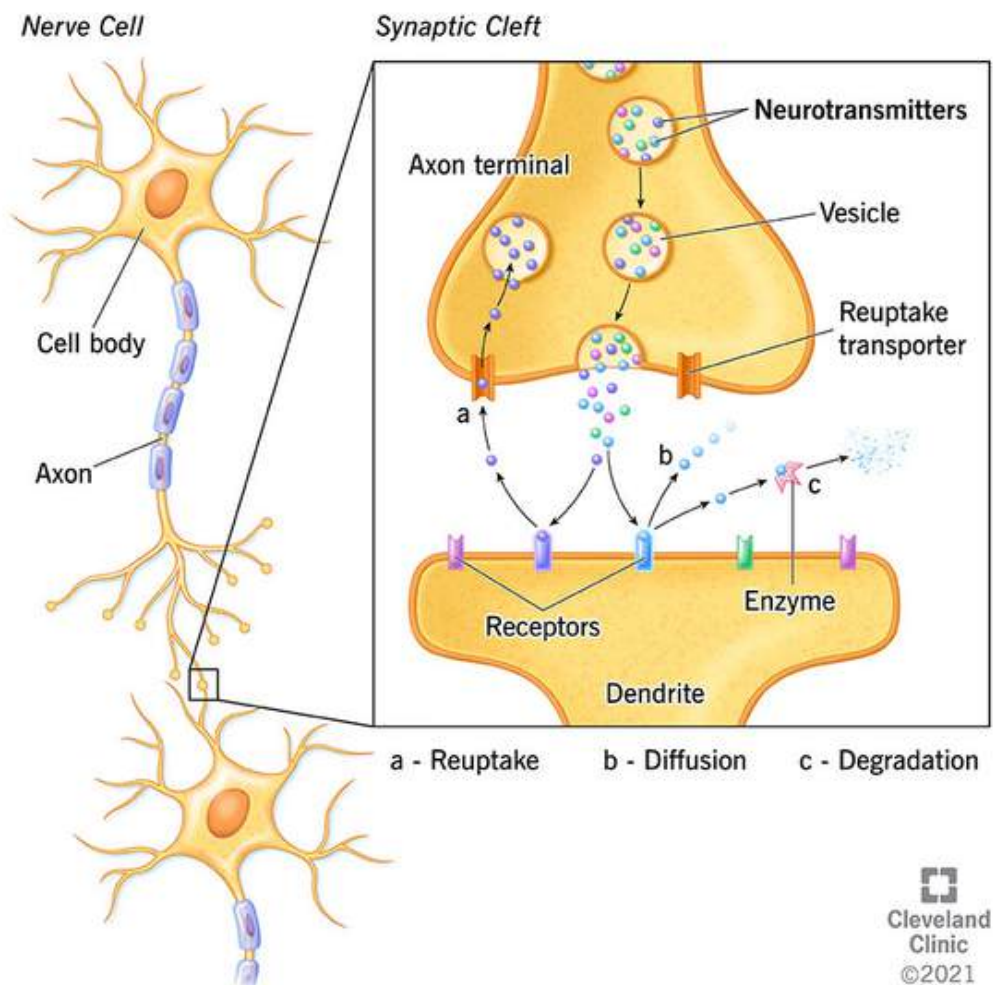
What is a Neurotransmitter?

BY COLIN KIM DOH

Neurotransmitters are chemical messengers that transmit signals between nerve cells, muscles, and glands, allowing the nervous system to control movement, thoughts, emotions, and vital body functions. When their levels or actions are imbalanced, communication between cells is disrupted, leading to neurological or mental health disorders. Neurotransmitters have been fundamental to medical research on brain activity and human behavior and mental disorders since their discovery. Scientists and healthcare professionals investigate the function of chemical messengers in nerve cell communication while they study how chemical imbalances lead to

disorders such as depression and anxiety and Parkinson's disease and epilepsy. The general public previously lacked knowledge about how neurotransmitters function in daily human activities which include breathing, heart rate, digestion, and movement. Many people know terms like "dopamine" or "serotonin," but they may not fully understand how these chemicals interact in complex networks.

Basically, the Neurotransmitters are chemical messengers who transmit signals through synapses. The chemicals control human movement and states and cognitive processes and essential body functions. Imbalances are linked to conditions such as depression, which results from serotonin



deficiency and Parkinson's disease which results from dopamine deficiency and anxiety disorders. Many medications function by changing how neurotransmitters operate.

Nowadays, neurotransmitter knowledge serves as essential base knowledge for psychology and neuroscience studies, medicine and healthcare careers, and interpreting how psychiatric medications work.

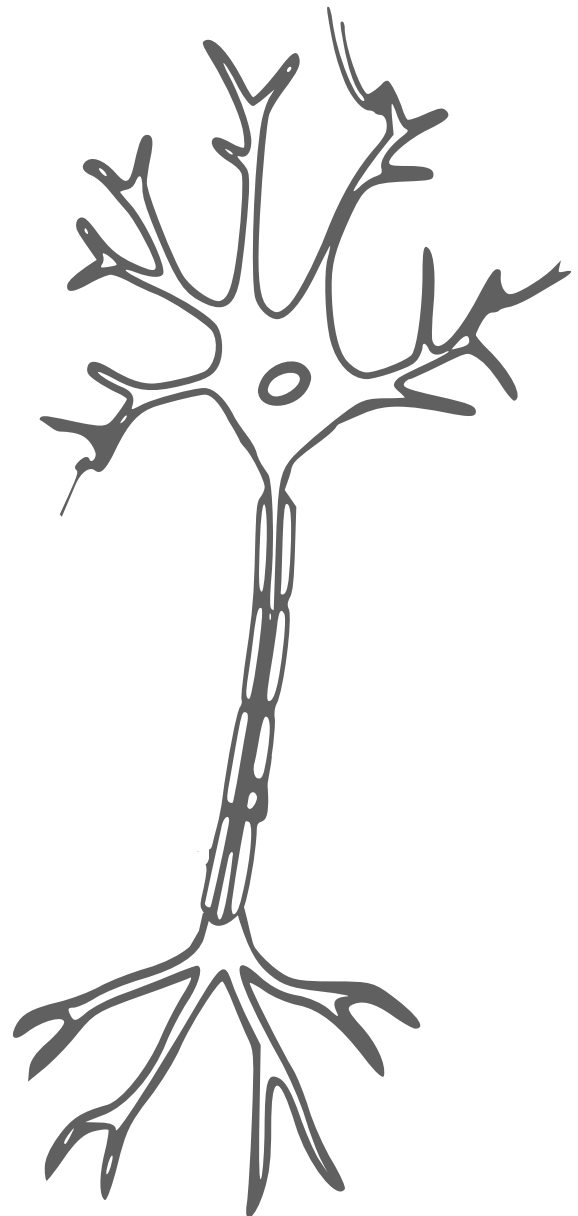
Biological systems require balance to function properly. Chemical signaling serves as the basis for both physical health and emotional health. Human behavior and disease development both depend on brain chemical interactions.

One particularly interesting aspect is that the same chemical messenger can have different effects depending on where it acts in the body. For example, dopamine plays a role in movement, motivation, and reward—showing how interconnected brain systems are.

By learning about neurotransmitters, patients can build a better understanding reduces stigma around mental health disorders, healthcare providers could enhance diagnosis and treatment strategies, researchers could continue and improve exploration of brain chemistry, and lastly, students could build foundational scientific literacy.

In the future, advances in neuroscience could lead to more precise medications with fewer side effects, early detection of neurological diseases, and improved mental health treatments.

Cleveland Clinic. (n.d.). *Neurotransmitters*. Cleveland Clinic.



(Canva)

Dyslexia, Learning and Memory

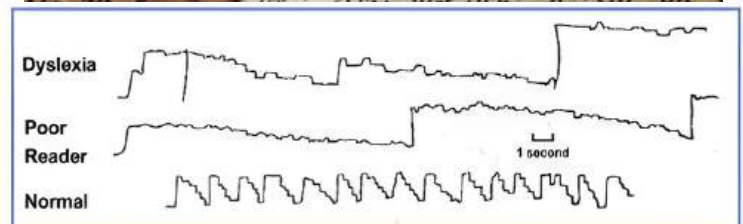
BY CHARLOTTE SHIN CHOE

For decades, the medical and educational communities viewed dyslexia through a narrow lens focused primarily on phonological processing. This traditional perspective suggested that the core issue was simply a struggle to map sounds to letters. While this remains a vital piece of the puzzle, recent neurological discussions have expanded to include the role of executive functions. Researchers are now investigating how the brain manages its internal resources, particularly attention and memory, during the complex task of reading (Sinha et al., 2024).

One of the most significant gaps in previous knowledge involved the specific behavior of the working memory network. While it was well known that children with dyslexia often struggle with memory tasks, scientists did not fully understand how these networks functioned during the precise moment of word decoding. It remained unclear whether these children were using the same brain regions as typical readers but with less efficiency, or if their brains were forced to recruit entirely different neurological pathways to compensate for their struggles.

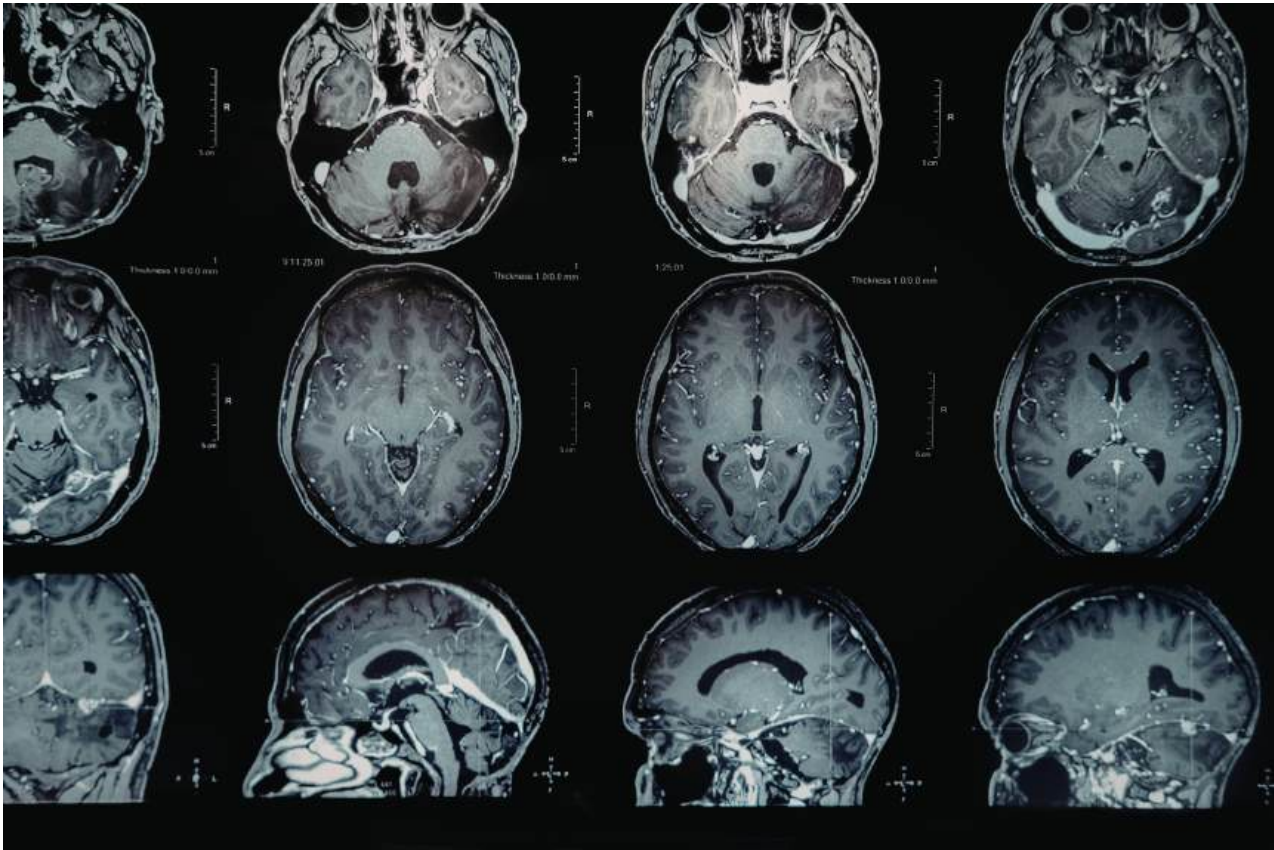
To understand these findings, the audience must first grasp the concept of working memory. According to foundational research, working memory acts as a mental scratchpad with limited space (Baddeley, 2000). In the context of reading, this scratchpad holds the first sound of a word while the brain works to figure out the last sound. If the scratchpad is too small or becomes cluttered, the child cannot successfully blend those sounds together to understand the word.

The study titled "The reading-attention relationship: Variations in working memory network activity during single word decoding in children with and without dyslexia" sought to map these mental scratchpads in real time. Conducted by a multi-institutional team from Georgia State University and the University of Connecticut, the research was published in the journal *Neuropsychologia* in 2024. The primary objective was to identify the neural signatures of dyslexia within the working memory system to understand why reading feels so much more physically and mentally exhausting for these students.



"Eye movements during reading. Up and down movements of the pen recorder correspond to left and right"

(Neuroscience, 2003)



(Canva)

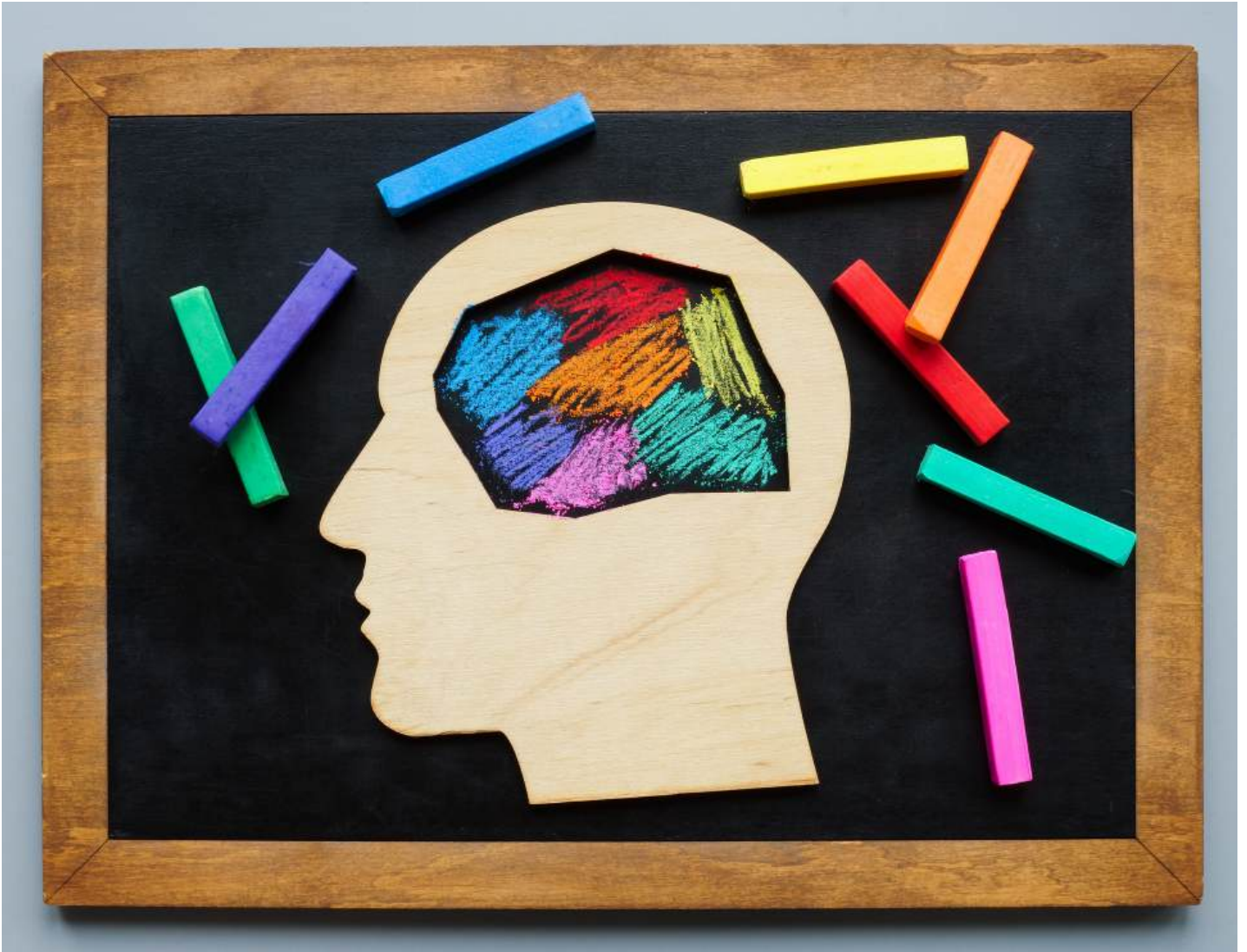
The researchers hypothesized that children with dyslexia would show significant over-activation in specific memory regions. They believed this extra activity would serve as a marker for a brain working harder to perform tasks that should be automatic for typical readers (Sinha et al., 2024). To test this, the team collected functional Magnetic Resonance Imaging (fMRI) data from ninety nine children. While inside the scanner, these children performed a memory task known as a 1-back test. They were shown three distinct types of visual information: real words, pseudowords, and false fonts.

By tracking oxygenated blood flow, the fMRI allowed the researchers to see exactly which parts of the brain were working during these tasks. The results were striking. While typical readers processed language primarily using the left side of the brain, children with dyslexia displayed bilateral recruitment. This means they were using both the left and right hemispheres to manage the memory load of the words. Despite this massive expenditure of brain power, the children with dyslexia still had lower accuracy and slower reaction times. This led the researchers to conclude that the dyslexic brain is significantly less efficient, using a sprawling and

exhausting network to do what a typical brain does with a small, specialized circuit.

These findings carry heavy implications for how we view student performance. One of the most important takeaways is that dyslexia is not a matter of a lack of effort. On the contrary, the data proves that the brains of students with dyslexia are often working double time compared to their peers. This provides a scientific explanation for cognitive fatigue. When a student uses both hemispheres of the brain just to decode a single sentence, they will naturally experience mental exhaustion much faster than a typical reader.

The study also highlighted a fascinating theme regarding the universality of this struggle. Because the differences in brain activity were visible even when the children looked at false fonts, it suggests that the working memory difference is a foundational part of how the brain is wired. It is not just a reading problem that appears when a book is opened; it is a fundamental difference in how information is filtered and stored. This extra brain activity creates a sort of internal traffic jam that makes it difficult for the child to focus on the actual meaning of the text.



(Canva)

The conclusion of this research offers a roadmap for the future of healthcare and education. For educators, the findings suggest a desperate need for low-load learning environments. Since the mental scratchpad of a student with dyslexia is easily overloaded, teachers should focus on providing one instruction at a time and breaking complex tasks into tiny, manageable steps. For parents, this research offers a sense of validation, explaining why a child might be able to read a word on one page but seem to forget it by the next. It is often a matter of how much memory space is available at that specific moment. In the medical field, these findings raise new questions about intervention.

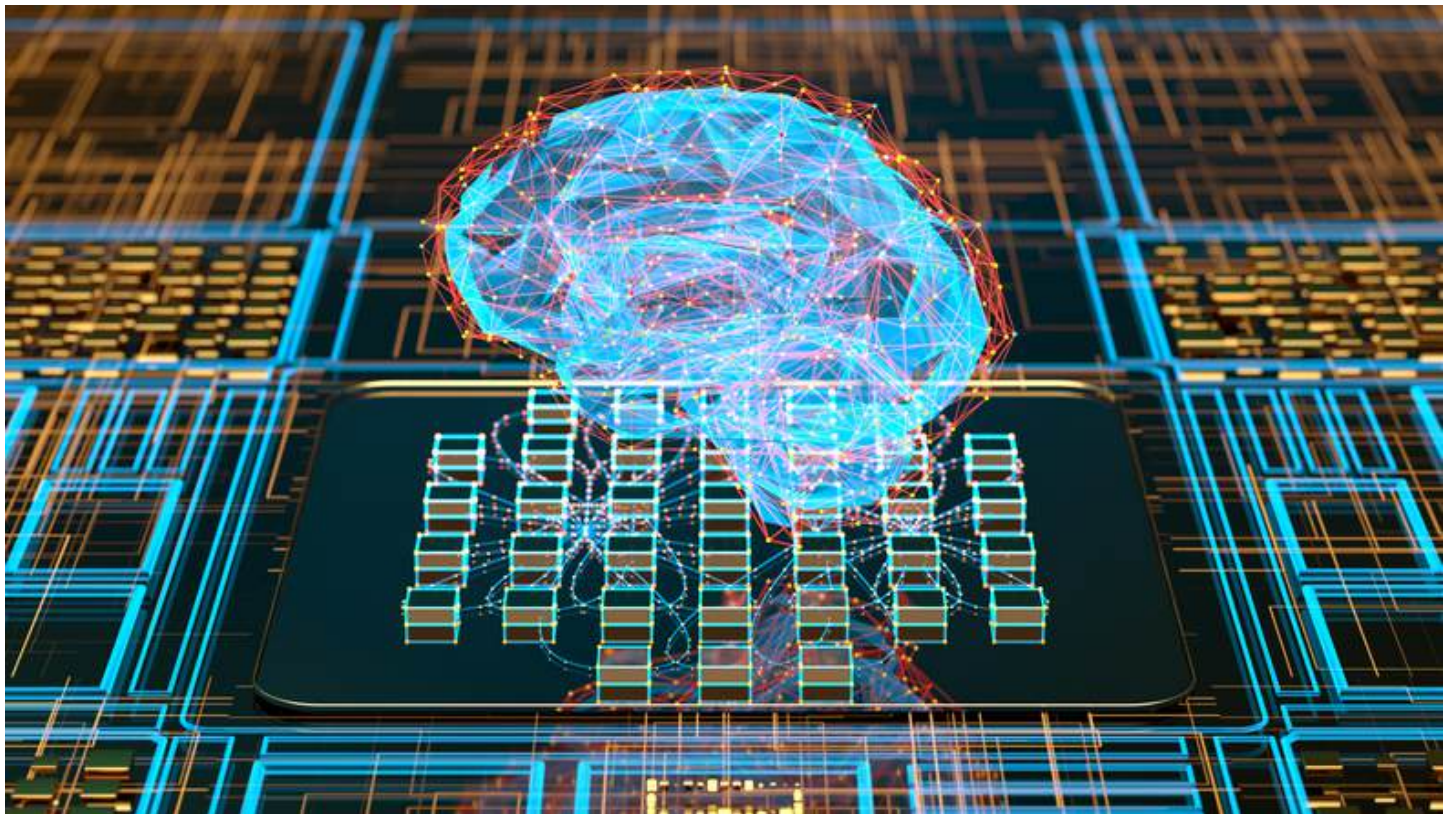
Scientists are now asking if we can train the brain to return to the more efficient left-hemisphere pathways or if we should instead find ways to make the right-hemisphere compensation more effective. There is also the potential for a massive shift in how we diagnose learning differences. In the future, fMRI scans could become a standard part of early childhood screenings. Instead of waiting for a child to fail a reading test in the second grade, doctors could identify these bilateral brain patterns in preschool. This would allow society to provide support and resources before the child ever has to face the emotional toll of academic failure.

Baddeley, A. D. (2000). The episodic buffer: A new component of working memory? *Trends in Cognitive Sciences*, 4(11), 417-423. [https://doi.org/10.1016/S1364-6613\(00\)01538-2](https://doi.org/10.1016/S1364-6613(00)01538-2)

Sinha, N., Arrington, C. N., Malins, J. G., Pugh, K. R., Frijters, J. C., & Morris, R. (2024). The reading-attention relationship: Variations in working memory network activity during single word decoding in children with and without dyslexia. *Neuropsychologia*, 193, 108764. <https://doi.org/10.1016/j.neuropsychologia.2023.108764>

Thinking Like a Brain: How Diffusive Memristors are Redefining Neuromorphic Computing

BY ROY BAEK



Introduction

Artificial intelligence has made progress in recent years, but the computers it runs on are still really inefficient. Modern computers use silicon-based transistors and electrons to process information, and today's big AI systems use megawatts of power to train on billions of data points.

Meanwhile, the human brain performs equivalent tasks with much less energy (roughly 20 watts, the energy of a dim light bulb). For a time, scientists and engineers have known that the brain achieves this efficiency through a mix of electrical and chemical signals. When an electrical impulse reaches the end of a neuron, it releases neurotransmitters across a junction called a synapse.

Once received by the adjacent cell, the chemical signal is converted back into a signal, which continues to move through the network. This mix of chemical processes helps the brain store, keep, and send information to our bodies.

The field of neuromorphic computing (designing chips that mimic the brain functions) has long sought to replicate these mechanisms in hardware. Earlier chips that tried to mimic the brain had brain cells that needed tens to hundreds of transistors each, which limited how big and efficient they could be. What remained unknown was whether researchers could build artificial neurons that physically recreate the ion-based dynamics of real biological neurons.

Findings

A research team led by Professor J. Joshua Yang of the Department of Computer and Electrical Engineering at the University of Southern California published the study that investigated whether such technology could come to life.

His team sought to demonstrate that ion motion in a solid-state device could imitate the fundamental neural mechanisms. Then, they studied ways to reduce the physical footprint of such a neuron to match that of a single transistor. Lastly, they also explored ways to effectively reduce energy consumption relative to existing neuromorphic models.

The researchers created a tiny artificial neuron using just three parts: a diffusive memristor, a transistor, and a resistor, occupying approximately 4 μm^2 per neuron. Inside the memristor, silver ions move through the oxide material in a way that mirrors how ions travel through channels in a real brain cell.

To test whether their artificial neurons behaved like real brain cells, the researchers sent electrical signals into the devices and carefully measured their responses. They observed three hallmark behaviors: the moment a neuron "decides" to fire (threshold-triggered firing), the brief recovery pause after firing (refractory period), and the way a neuron slows its firing rate over time when repeatedly stimulated (frequency adaptation). Finally, they tracked how silver particles inside the device moved to confirm that their motion closely mirrors the motion of ions in a real biological neuron.

As a result, the researchers confirmed that their diffusive memristor-based neurons successfully replicated multiple key properties of biological neurons using actual ion motion rather than mathematical simulation. Each artificial neuron fits within the footprint of a single transistor, compared to tens or hundreds of transistors required by conventional neuromorphic designs. This represents a potential reduction in chip size and energy consumption.

Conclusion

One aspect of the findings that really stands out is how simple the solution is. By making complicated circuits with many parts to mimic how neurons work, the team discovered that a single part that allows ions to move through it can show the same complex behavior as real neurons. This discovery has a lot of potential for designing intelligent hardware in the future.

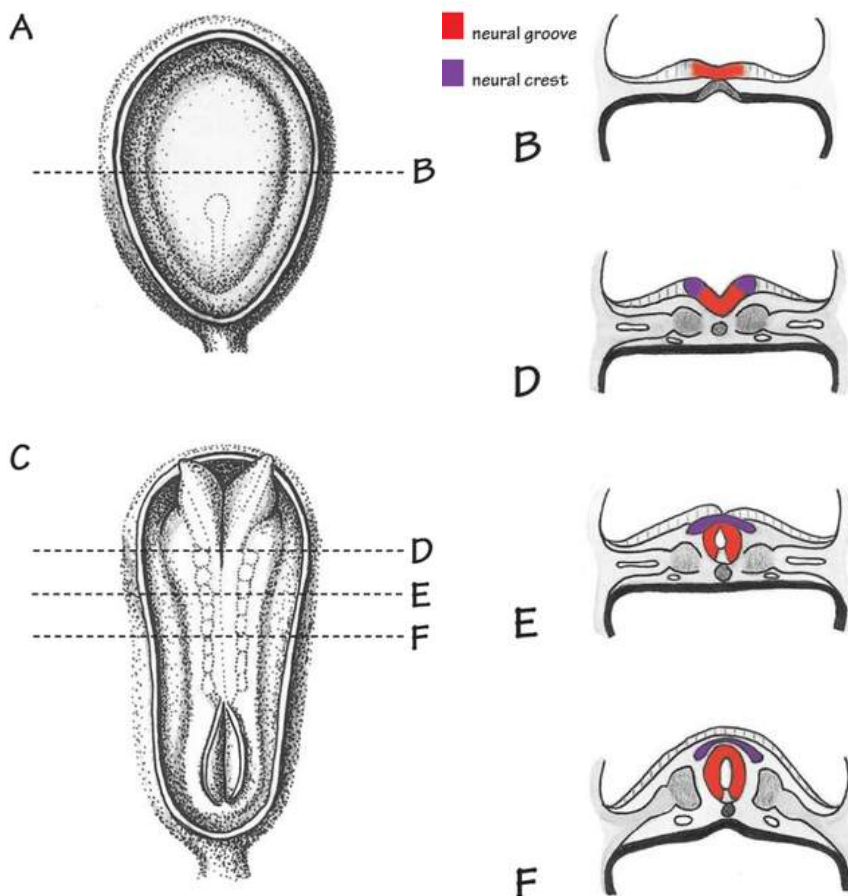
For AI researchers and chip manufacturers, these findings can allow the development of more innovative designs with flexible shapes and efficient energy consumption. For clinicians and healthcare institutions, more efficient AI hardware could eventually enable diagnostic and monitoring systems that operate continuously on minimal power, especially in low-resource settings. For policymakers, the prospect of reducing AI energy consumption addresses growing concerns about the carbon footprint of machine learning infrastructure. For neuroscientists, an artificial system that physically mimics neural ion dynamics provides a new experimental platform for testing theories of brain function.

Some of the remaining tasks should still be addressed by future studies. It is important to make sure the silver used in the experiment is compatible with standard semiconductor manufacturing processes. Future studies can look into materials that can have similar effects. In addition, the more we can make systems that work like the brain, the more we can learn about how the brain really works.

Zhao, R., Wang, T., Moon, T. et al. A spiking artificial neuron based on one diffusive memristor, one transistor and one resistor. *Nat Electron* 8, 1211–1221 (2025). <https://doi.org/10.1038/s41928-025-01488-x>

Brain Development: From Embryo to Fetus

BY DANU KIM

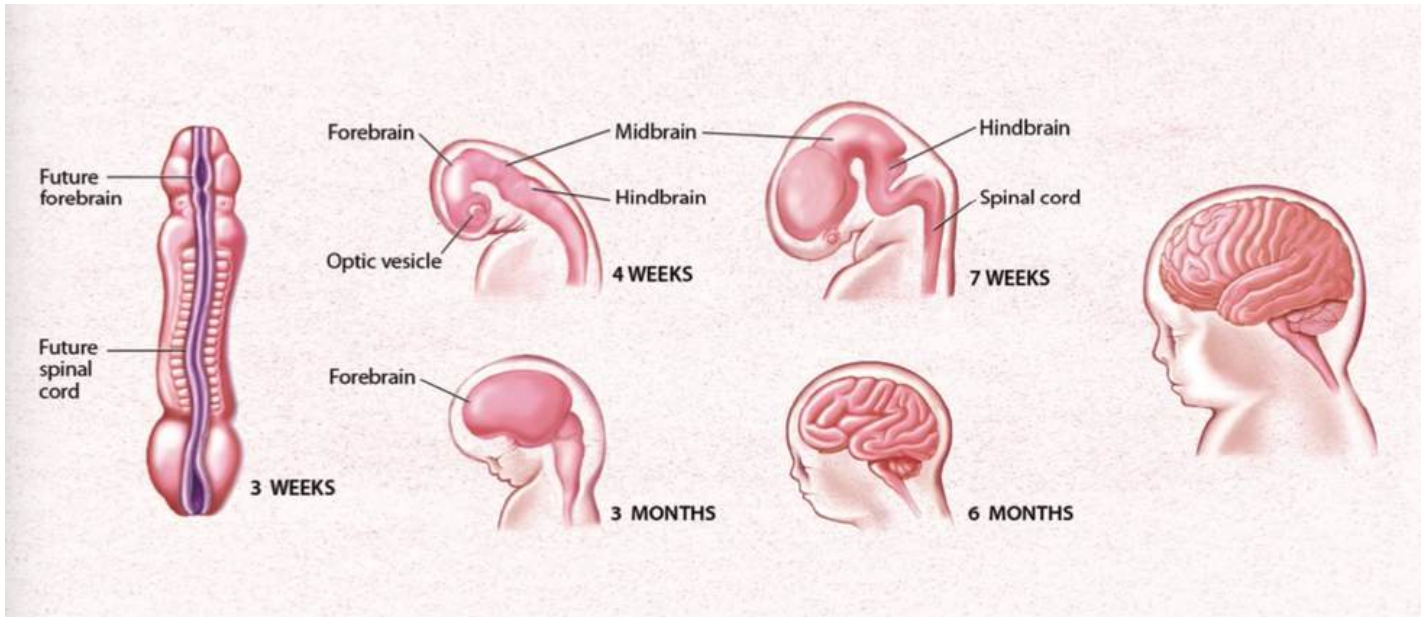


(Neuroscience, 2003)

In the embryo stage, the first step in our brain development is cell division. Another key step is cell differentiation, in which individual cells stop dividing and take on specific characteristics, such as neurons or ganglion cells, where the ganglion cells are another type of neural cell that assist the neurons. Differentiation orders the things spatially, where the different kinds of neurons migrate to various locations in a process called pattern formation.

The first major event of pattern formation takes place in the third week of human gestation, when the embryo is just two connected sheets of dividing cells. A small patch of cells on the upper surface of the bilayer is instructed to make the entire brain and spinal cord. These cells form a tennis racket-shaped structure called the neural plate, the front of which is destined to form the brain, and the rear to be the spinal cord. Signals directing the destiny of these cells come from the layer beneath that goes on to form the midline skeleton and muscles of the embryo, which is called mesoderm. Various regions of the early nervous system express

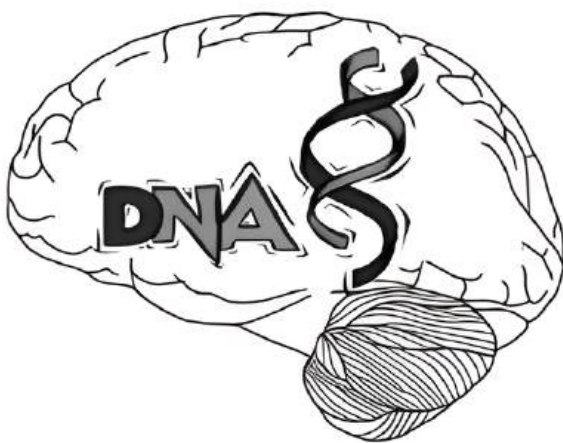
different subsets of genes preceding the emergence of brain areas: forebrain, midbrain, and hindbrain. In the above illustration, "A" represents the human embryo at 3 weeks after conception, which we just talked about. "B" talks about the neural plate forming the top surface of the embryo. "C" displays the embryo a few days later, when the embryo develops enlarged head folds at the front. "D," "E," and "F" illustrate the different levels of the axis from head to tail, showing various stages and neural tube closure. So a week later, the neural plate rolls up, closes into a tube (F), and sinks into the embryo. In other words, initially the neural tube develops with the genes of our skin cells, and eventually the neural tube, or the future spinal cord and brain part, sinks into our skin. So it becomes enveloped by the future epidermis, which is another word for our skin. Further changes will happen in



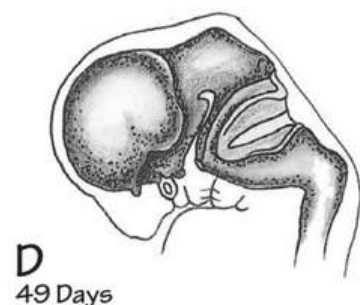
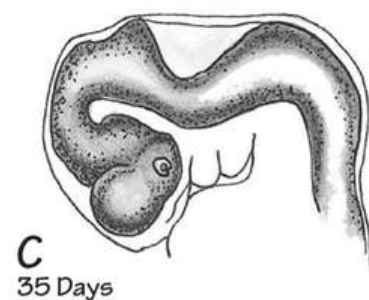
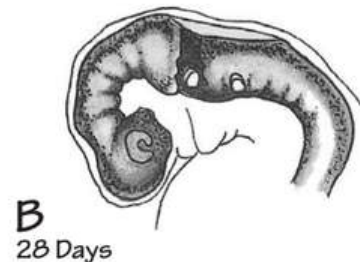
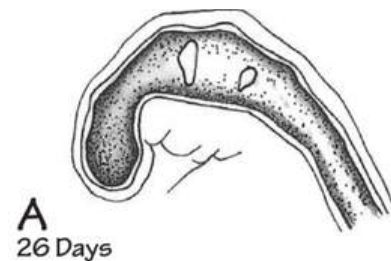
(BrainFacts, 2012)

the next few weeks, such as changes in cell shape, division, migration, and cell adhesion. For example, the neural tube flexes such that the head region is bent at right angles to the trunk region. This patterning process leads to finer levels of resolution, ultimately conferring individual identity to the young neurons.

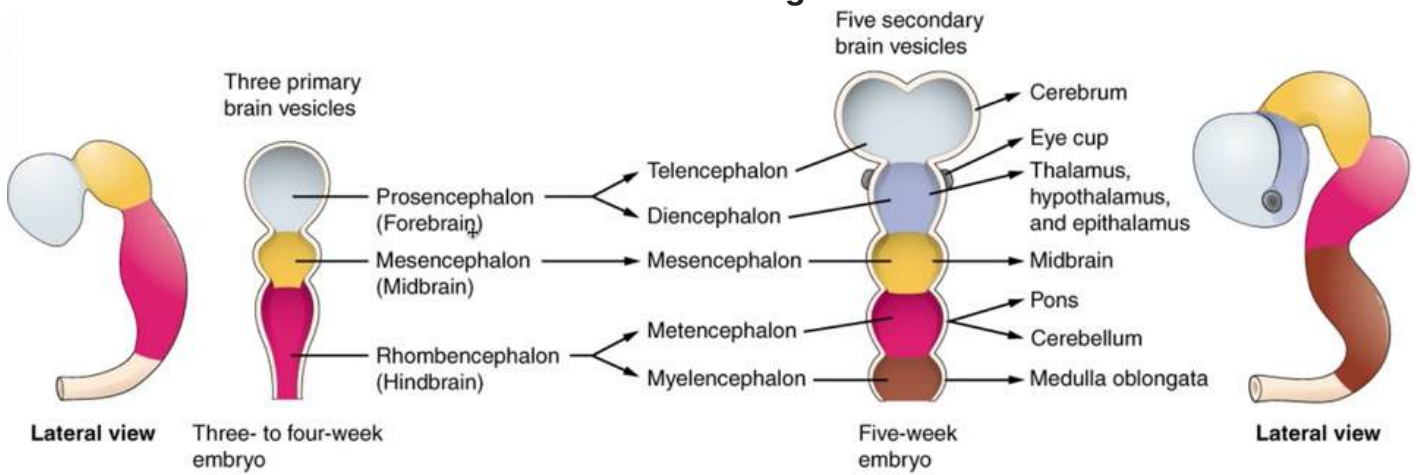
The above image is another model showing the different parts of the baby's brain and nervous system. There are the forebrain, midbrain, and hindbrain. And as the time goes on, to 3 weeks, 4 weeks, 3 months, and 6 months, the brain begins to form, and it begins to look like the big walnut that we know now.



(Neuroscience, 2003)



(Neuroscience, 2003)



(National Center for Biotechnology Information)

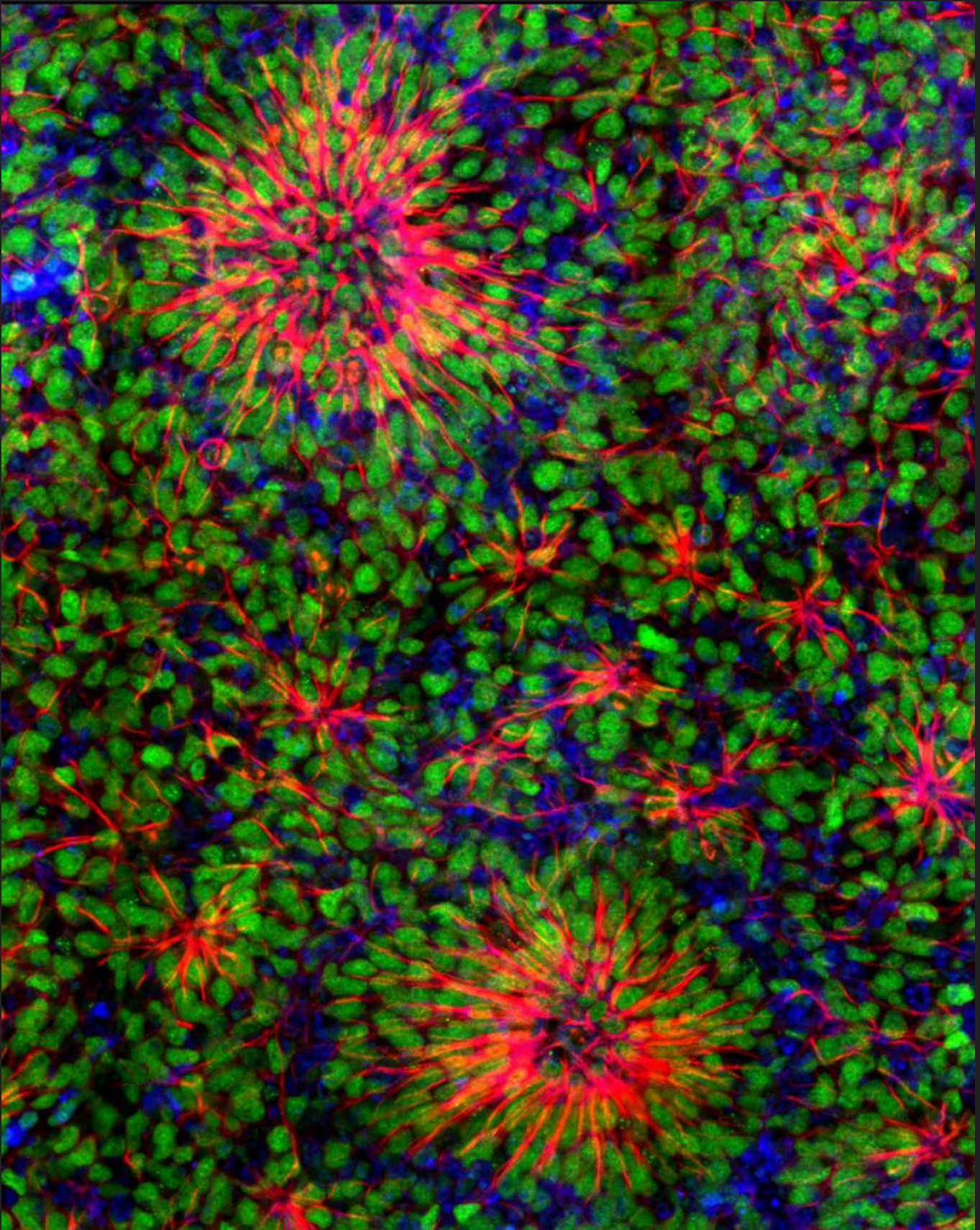
Now let's talk about the different names of brain parts from their initial development stage to the growing stage. Initially, in the 3 to 4-week embryo that we just talked about in the previous section, the brain part that looks like illustration "A," there are three primary brain parts. "Cephalis" is a Greek word for "head," and thus "Cephalon" is used to identify the different brain parts. There are three primary brain parts in the initial stage from 3 weeks to 4 weeks of the embryo. There are the Prosencephalon, Mesencephalon, and Rhombencephalon. The prosencephalon got its name from the fact that it is at the very beginning, at the very front of the brain, as you can see in the lateral view in the above image. Because "pro" means "in the very front," it is named that way. Mesencephalon got its name from the fact that "meso" means "mid" and the mesencephalon is in the middle of the growing brain. Rhombencephalon got its name from the fact that it would look like a rhombus-like structure when it develops later. Now, the Prosencephalon divides into two sections, the Telecephalon and the Diencephalon. "Tele" means far away, so the fact that that part of the brain is far away from the rest of the brain led to the name Telecephalon.

"Di" means two or second, and Diencephalon is the second part of the brain, which is where Diencephalon got its name. Mesencephalon is the same as before. Rhombencephalon divides into Metencephalon and Myelencephalon. Metencephalon got its name from the root word "meta," which means "change." "Meta" plus "Cephalon" became Metencephalon, and "Myelencephalon" got its name from the fact that it later develops to protect the spinal cord. "Myelin" is a word that represents "protection." These different initial brain parts develop into the actual brain parts like cerebrum, eyes, thalamus, hypothalamus, epialamus, midbrain, pons, cerebellum, and medulla oblongata, and they later develop into a complex nervous system with neurons that communicate with each other to let humans survive and live well.

British Neuroscience Association & European Dana Alliance for the Brain. (2003). Neuroscience: Science of the brain—An introduction for young students.

Society for Neuroscience. (2012). Brain facts: A primer on the brain and nervous system (7th ed.). Society for Neuroscience.

National Center for Biotechnology Information. (n.d.). Figure 1 [Image]. In NCBI Bookshelf. U.S. National Library of Medicine. <https://www.ncbi.nlm.nih.gov/books/NBK526024/figure/article-19131.image.f1/>



Medicine
